Experiment No. 03:

Experiment Title: Detection of the light using photo resistor(LDR).

Objectives

- Learn how to interface a Light Dependent Resistor (LDR) with an Arduino Uno.
- Understand light intensity detection using a voltage divider circuit.
- Read analog light intensity values using the Arduino's 10-bit Analog-to-Digital Converter (ADC).
- Program the microcontroller to trigger an LED output based on a predefined ambient light threshold.

Introduction

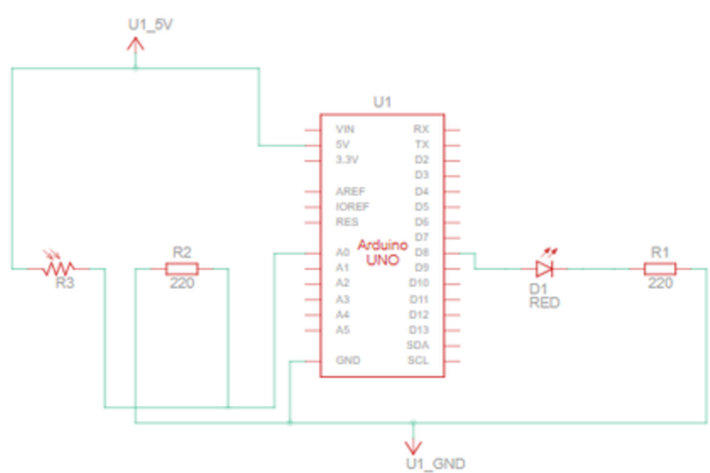
A Light Dependent Resistor (LDR), or photoresistor, is a passive sensor whose electrical resistance decreases as the intensity of light falling on it increases. In complete darkness, an LDR exhibits very high resistance , whereas in bright light, its resistance drops to a few hundred ohms (Omega).

- Voltage Divider Network: Because microcontrollers read voltage rather than raw resistance, the LDR is paired with a fixed 10kΩ resistor to form a voltage divider. The output voltage (V out) varies proportionally with light levels.
- Analog Reading & Threshold Logic: The variable output voltage is fed into Analog Pin A0, where the ATmega328P ADC converts 0V - 5V into digital values ranging from 0 to 1023. The microcontroller evaluates this value against a code threshold to automatically turn an LED ON (e.g., when dark) or OFF (e.g., when light).

Apparatus Required

Component / Equipment	Quantity	Specification / Description
Arduino Uno R3	1	Microcontroller development board
LDR (Photoresistor)	1	Light-sensitive variable resistor
LED	1	Standard 5mm Light Emitting Diode
Resistor (10kΩ)	1	Fixed resistor for voltage divider network
Resistor (220Ω)	1	Current-limiting resistor for the LED
Breadboard	1	Solderless prototyping board
Jumper Wires	As needed	Male-to-Male jumper wires

Circuit Diagram



Working Procedure

1. Drag and place the Arduino Uno, LDR, LED, 10kΩ Resistor, and 220Ω Resistor onto the Tinkercad workspace.
2. Connect the cathode (short leg) of the LED directly to the Arduino's GND pin.
3. Connect the anode (long leg) of the LED to a 220Ω resistor, and connect the other end of the resistor to Digital Pin 13.
4. Connect one terminal of the LDR to the 5V power pin on the Arduino.
5. Connect the second terminal of the LDR to Analog Pin A0.
6. Connect a 10kΩ fixed resistor from the junction between the LDR's second terminal and Pin A0 down to the GND rail to complete the voltage divider.
7. Open the Tinkercad code window, select C++ text mode, and paste the source code.
8. Click Start Simulation, open the Serial Monitor, select the LDR in Tinkercad, and adjust the ambient light slider to observe sensor readings and automatic LED switching.

Code

```
int sensorPin = A0;
int ledPin = 8;
int lightVal;
void setup(){
pinMode(ledPin, OUTPUT);
Serial.begin(9600);}
void loop() {
lightVal = analogRead(sensorPin);
Serial.println(lightVal);
if(lightVal < 100) {
digitalWrite(ledPin,HIGH);
}
else {
  digitalWrite(ledPin, LOW);}}
```

Discussion and Conclusion

- Discussion: The circuit successfully detected light intensity variations using the LDR and voltage divider combination. When ambient light increased, the LDR's resistance decreased, raising the voltage at pin A0 towards 5V (producing higher ADC readings). Conversely, covering the LDR increased its resistance, dropping the voltage at pin A0 (lower ADC readings). By setting a calibrated threshold (e.g., 500), the Arduino functioned as an automatic light detector, toggling the LED when light dropped below the set point.
- Final Verdict: The experiment was completed successfully. The Arduino accurately processed light level changes detected by the photoresistor and executed automated output control based on threshold logic.